مدارس الإمارات الوطنية
Emirates National Schools

Grade: 6 | Term 1, Week 10 (2 Nov to 6 Nov) | TCD: Future Skills 6: Applied Technology Foundations | Session length: 2 hrs

T1 WEEK: 10	**Lesson 1: Laser Cutting My Maker Mark**
Standard(s):	Math: 6.G.A (area and dimensions of shapes), 6.RP (scale); Technology: 2D CAD to fabrication workflow; Science: material properties
Literacy Strategies:	Laser vocabulary cards (vector, raster, cut, engrave, kerf, power, speed); frame: The laser will cut ... and engrave ...
Assessment:	Laser-ready file checked by the teacher; observation of safety routines; live quiz on the TCD website.
Lesson Objectives:	We are learning to prepare a 2D design for the laser cutter and to understand how the laser cuts and engraves.
Success Criteria:	I can explain the difference between cutting and engraving. I can set up a file with a cut outline and an engrave area. I can choose a suitable size and material for my piece. I can follow the laser cutter safety rules.
Targeted Vocabulary	laser cutter, vector, raster, cut, engrave, kerf, plywood, acrylic, power, speed
Resources	Laser cutter (teacher operated) with extraction running, 3 mm plywood or acrylic sheets, Computers with laser software or a vector editor, Students' maker mark files, Keyrings, Safety glasses, Sample cut and engraved pieces Accompanying resource: Grade6_Week10_Presentation.pptx (slides, hook video and live quiz code on the TCD website), Grade6_Week10_Worksheet and Grade6_Week10_Lesson_Plan.
National Identity Integration	Link to laser-cut mashrabiya patterns in modern UAE architecture, such as the shading screens on buildings in Abu Dhabi and Dubai.
AI/Digital Learning	Laser software on the teacher's computer (for example LightBurn or the laser's own software); Canva files from Week 9; hook video; live quiz on the TCD website.
Direct Instruction: I Do	Explain how the laser works (slide 6). Model setting cut and engrave layers on a sample file (slide 7). Explain size, material and kerf (slide 8). Go through laser safety (slides 9 and 11).
Guided Practice: We Do	Students import their maker mark and draw the outline with teacher support (slide 10). Partners check each other's files against the checklist.
Independent Practice: You Do	Students finalise their laser file and get the teacher's check (slide 12). The teacher runs the laser in batches while students observe from behind the safety line and record settings (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: Beginning / On / Above	**Beginning: Use the ready-made circle keychain outline and place your maker mark inside.** **On: Design your own outline shape with a hole and correct layers.** **Above: Design a plaque with your maker mark and a short engraved motto, and calculate how many pieces fit on one sheet.**
Closure/Reflection	Students compare finished pieces with their file. What changed? Exit ticket (slide 15). Pieces are kept for finishing next week.
Inclusion / EAL Support	☑ Visual Supports ☑ Chunked Instructions ☐ Reduced Language Load ☑ Model / Worked Example ☑ Pre-teach Vocabulary ☐ TA/LSA Support ☐ Other: ☐ Extra Processing Time ☐ Text Read by TA/LSA ☐ Flexible Grouping ☐ Assistive Technology ☐ Movement / Sensory Break ☐ Scaffold / Sentence Starters

مدارس الإمارات الوطنية
Emirates National Schools

G&T / Extension	☑ Extension / Challenge Task ☑ Real-World Application ☐ Higher-Order Questioning ☐ Curriculum Compacting ☑ Depth & Complexity Task ☐ Above-Grade-Level Texts ☐ Independent Inquiry ☐ Student-Led /Passion project ☐ Leadership / Peer Coaching ☐ Open-Ended Task ☐ Other: _____